Fpt University

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CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Ho Chi Minh City, November 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
09/11/2026	A	Trương Nguyễn Thái Bình (SE182370)	Initial software test documentation: scope of testing, test strategy, test plan, unit test cases (38 functions) and integration / system test cases (45 modules) with their statistics.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

Table 1: Scope of Testing

Feature	Function	Role	Description
Authentication	Sign Up	Guest	Ensure the guest can create a CUSTOMER account with valid information or by phone OTP, and the system shows an error for invalid inputs.
	Sign In with Password	All Roles	Check that the user can sign in with email / phone and password and is taken to the home page of the account's role.
	Sign In with Phone OTP or Google	Guest, Customer	Ensure the Firebase phone OTP and Google sign-in are verified on the server and wrong codes are rejected.
	Sign In with Email OTP	All Roles	Ensure the 6-digit email OTP is accepted when correct and rejected when wrong or expired (300 s).
	Reset Password	All Roles	Test the password-reset flow: request the OTP by email, verify it and set a new password.
	Admin Login with 2FA	Admin	Ensure the admin needs the password and the email OTP, and the session is closed after 5 wrong OTPs.
	Log in to Kiosk by QR	Customer	Ensure the customer can scan the kiosk QR (valid 2 minutes) and confirm, and expired or reused sessions are refused.
Profile Management	Update Profile	All Roles	Ensure all users can update their personal information and avatar.
	Change Password	All Roles	Verify a registered user can change the password and the other sessions are revoked.
	Delete Account	Customer	Ensure the account is deleted, its email / phone are released and it can no longer sign in.
Store & Locker Search	Search Stores and Lockers	Guest, Customer	Ensure users can search stores and lockers on the list and the map with online status and free boxes.
	View Locker Box Map	Customer	Check that the real box layout shows the state of every box and only free boxes can be booked.
Locker Send	Send Parcel	Customer	Ensure the customer can create a SEND order with receiver, voucher and points, and the box is held until payment.
	Drop Parcel and Share Pickup Code	Customer	Ensure the sender can open the paid box, confirm the drop and share the pickup code; the receiver receives the code.

Feature	Function	Role	Description
	Change PIN and Delegate Pickup	Customer	Ensure the sender can regenerate the PIN or delegate the pickup and the old PIN stops working.
	Pick Up Parcel at Kiosk	Receiver	Ensure the receiver opens the box with the PIN or QR, and the box is locked after 5 wrong attempts.
Locker Rental	Rent Locker	Customer	Ensure the customer can rent a box by the hour, pay and reopen it with the rental PIN.
	Extend and End Rental	Customer	Ensure the customer can extend a rental (only the difference is charged) and end it.
	Pay Overdue Fee	Customer, Receiver	Ensure the box opens only after the overtime fee is paid and overdue orders expire after 72 hours.
Payment	Pay by VietQR (SePay)	Customer	Ensure the order becomes PAID only when the SePay webhook confirms the full amount, and duplicated webhooks are ignored.
	Pay with Loyalty Points	Customer	Ensure point redemption follows the limits (at least 1,000 points, at most 50 % of the order) and the rest is paid by VietQR.
	Request Refund	Customer	Ensure the customer can request a refund within 7 days with a reason and follow its status.
Order Management	View My Orders and Incoming Orders	Customer, Receiver	Ensure the customer and the receiver see only their own orders with status history and payments.
	Rate Order and Send Complaint	Customer	Ensure the customer can rate a completed order once or send a complaint and receive the admin reply.
Promotion & Loyalty	Save and Apply Vouchers	Customer	Ensure vouchers can be saved, applied with their scope and minimum amount, and exchanged for points.
	Earn Loyalty Points	Customer	Ensure completed orders earn 10 % as points, tiers follow lifetime points and points are reversed on refund.
Incident Report	Report Locker Incident	Customer	Ensure the customer can report a faulty box with title and photos and the box becomes PENDING_CHECK.
Drone Delivery	Book Drone Delivery	Customer	Ensure booking checks the receiver locker, weight (<= 5,000 g), distance (<= 5 km), weather and calculates the fee by weight.
	Track Drone Delivery	Customer, Receiver	Ensure the customer sees every stage, the live position and can cancel only while awaiting dispatch.
	Drone Home Pickup	Customer, Drone Technician	Ensure the pickup is offered to the nearest technician on duty and fails with a refund when not served in time.

Feature	Function	Role	Description
Drone Operation	Dispatch and Confirm Loading	Drone Technician	Ensure the technician accepts an order with an idle drone (battery >= 20 %) and confirms weight, seal code and checklist.
	Launch and Report Flight Failure	Drone Technician	Ensure launch respects flight hours and suspensions, and a failed flight puts the order in refund.
	Manage Drone Fleet	Drone Technician	Ensure the technician can claim drones, change status with reason and update battery and maintenance notes.
Maintenance	Manage Shift	Locker Technician, Drone Technician	Ensure only technicians on shift with location on receive tickets, offers and missions.
	Handle Incident Ticket	Locker Technician	Ensure the technician can claim, extend the SLA (max 2 x 24 h), return and resolve tickets with photos.
	Perform Periodic Inspection	Locker Technician	Ensure passing all items advances the due date and a failed item opens a ticket automatically.
	Emergency Open and IoT Devices	Locker Technician	Ensure the technician can open boxes, control the lid and restart controllers only of assigned lockers.
Administration	View Dashboard and Revenue Report	Admin	Check that the dashboard and the revenue report use the money actually collected and refunds are deducted.
	Manage Accounts	Admin	Ensure the admin can search users, create staff, lock / unlock and delete accounts.
	Manage Stores, Lockers and Boxes	Admin	Test CRUD of stores, lockers and boxes, batch creation and that occupied boxes cannot be deleted.
	Manage Drones and Drop Incidents	Admin	Ensure the admin can register drones and decide free redelivery or compensation for dropped parcels.
	Process Refunds	Admin	Ensure the admin approves with a transfer receipt or rejects with a reason.
	Manage Promotions and Loyalty Rules	Admin	Ensure the admin can create promotions with period, usage limit, locker scope and point price.
	Configure Business Settings and Jobs	Admin	Ensure business rules can be changed without restart, are audited, and scheduled jobs can be run on demand.
Notification	Receive Notifications	All Roles	Ensure push and in-app notifications are delivered in Vietnamese and only the owner can read or delete them.
AI Assistant	Ask AI Assistant	All Roles	Ensure answers cite only documents the role may read and the asker's own orders, and off-topic questions are refused.

Feature	Function	Role	Description
	Manage Knowledge Base	Admin	Ensure uploaded documents are indexed with their allowed roles and the retrieval evaluation can be run.
IoT Locker	Open Box via Bluetooth	Customer	Ensure only granted accounts get a one-time signed token valid for 120 s to open a box over Bluetooth.
	Execute Cabinet Command	Locker Controller	Ensure the Pi executes open / lid / restart commands over MQTT and reports heartbeats, door states and results.

2. Test Strategy

2.1 Testing Types

Table 2.1: Testing Types

Objective	Technique	Completion criteria
Verify and validate each individual code unit (service methods, controllers, listeners) in isolation with mocked dependencies.	Unit testing	- All unit tests pass in CI on every pull request. - Every main service class of the backend is covered by tests.
Check the requests, responses and error codes exchanged through the REST API, the API gateway rules and the MQTT / STOMP message contracts.	Interface testing	- Requests and responses match the API contract (status codes, error codes, payload fields).
Check correct interaction among services, the PostgreSQL databases, RabbitMQ events and the payment webhook.	Integration testing	- Data passed between services matches the expected schemas. - Repository queries pass on a real PostgreSQL database.
Test the five business flows end to end on the deployed system with the mobile app, kiosk, admin web, cabinet and drone.	End-to-end testing	- Every test case of the test report is executed in round 1. - No open defect blocks a business flow.
Check that the Raspberry Pi controller drives relays, door sensors and the sliding lid of the real cabinet correctly.	Hardware testing	- Every box of the cabinet opens on command and reports its door state.

2.2 Test Levels

Table 2.2: Test Levels

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
Unit testing	X			
Interface testing	X	X		
Integration testing		X		
End-to-end testing			X	X
Hardware testing		X	X	

2.3 Supporting Tools

Table 2.3: Supporting Tools

Purpose	Tool	Vendor/In-house	Version
Unit testing (backend)	JUnit 5, Mockito, AssertJ	Open Source	JUnit 5.12.2, Mockito 5.17.0, AssertJ 3.27.7
Integration testing (database)	Testcontainers (PostgreSQL)	Open Source	1.21.4
Build and test runner	Apache Maven, GitHub Actions	Open Source / GitHub	Maven 3.9.10
API testing	Postman / Swagger UI (springdoc)	Open Source	Latest stable
Mobile app testing	flutter_test	Google (Open Source)	Flutter 3.44
Cabinet hardware testing	Pi service panel (/service)	In-house	1.0

3. Test Plan

3.1 Human Resources

Table 3.1: Human Resources

Worker/Doer	Role	Specific Responsibilities/Comments
Nguyễn Quốc Bảo Huy	Leader	Plan the testing, assign test functions to members, review test cases and verify test results.
Trương Nguyễn Thái Bình	Member	Write and execute the test cases of accounts, locker send / rental, payment and admin web; prepare the test report.
Lê Thị Yến Vi	Member	Write and execute the test cases of drone delivery and drone operation; write unit tests and report defects.
Trần Kim Nhã	Member	Write and execute the test cases of technicians, kiosk, IoT devices and the cabinet controller; report defects.

3.2 Test Environment

Table 3.2: Test Environment

Purpose	Tool	Provider	Version
Back-end server	Azure virtual machine + Docker Compose	Microsoft Azure	Ubuntu 24.04
Database	PostgreSQL, pgvector	PostgreSQL Global Development Group	16
Message broker	RabbitMQ, Eclipse Mosquitto (MQTT)	Broadcom, Eclipse Foundation	RabbitMQ 3, Mosquitto 2
Admin web testing	Google Chrome	Google	Latest stable
Mobile app testing	Android phone	Google	Android 11 or later
Locker cabinet testing	Raspberry Pi 5 with kiosk touch screen	Raspberry Pi Ltd	Raspberry Pi OS (64-bit)
Unit test execution	Windows 11 PC, JDK	Microsoft, Oracle	Windows 11, JDK 21

3.3 Test Milestones

Table 3.3: Test Milestones

Milestone Task	Start Date	End Date
Unit test	13/09/2026	09/11/2026
Other Test Case (IT, ST, AT)	28/09/2026	07/11/2026

4. Test Cases

Unit Test Cases: Report5_Unit Test_LockR.xls

UNIT TEST REPORT

Project Name	*Lock.R - Smart Locker System with Drone Deli*	**Creator**	**binhtntse182370**
Project Code	*LockR*	**Reviewer/Approver**	**huynqbse180211**
Document Code	*LockR_Unit Test_v1.0*	**Issue Date**	*09/11/2026*
Notes	*Release 1 includes 38 functions of 11 backend services (auth, user, store, order, payment, loyalty, notification, loc*		

No	Function code	Passed	Failed	Untested	N	A	B	Total Test Cases
1	F001	9	0	0	5	3	1	9
2	F002	5	0	0	3	0	2	5
3	F003	15	0	0	4	6	5	15
4	F004	15	0	0	3	6	6	15
5	F005	7	0	0	3	2	2	7
6	F006	11	0	0	6	4	1	11
7	F007	10	0	0	5	3	2	10
8	F008	10	0	0	5	3	2	10
9	F009	8	0	0	3	2	3	8
10	F010	13	0	0	5	4	4	13
11	F011	8	0	0	5	1	2	8
12	F012	7	0	0	3	3	1	7
13	F013	14	0	0	3	4	7	14
14	F014	8	0	0	4	4	0	8
15	F015	9	0	0	2	4	3	9
16	F016	13	0	0	4	7	2	13
17	F017	9	0	0	3	6	0	9
18	F018	11	0	0	5	3	3	11
19	F019	16	0	0	8	4	4	16
20	F020	14	0	0	6	4	4	14
21	F021	9	0	0	6	2	1	9
22	F022	12	0	0	2	2	8	12
23	F023	8	0	0	3	3	2	8
24	F024	7	0	0	3	4	0	7
25	F025	18	0	0	11	4	3	18
26	F026	10	0	0	4	4	2	10
27	F027	14	0	0	9	3	2	14
28	F028	13	0	0	6	5	2	13
29	F029	9	0	0	5	4	0	9
30	F030	12	0	0	2	7	3	12
31	F031	25	0	0	7	14	4	25
32	F032	16	0	0	5	9	2	16
33	F033	14	0	0	10	1	3	14
34	F034	6	0	0	2	2	2	6
35	F035	11	0	0	3	4	4	11
36	F036	8	0	0	4	2	2	8
37	F037	10	0	0	5	3	2	10
38	F038	0	0	14	5	5	4	14
	Sub total	414	0	14	177	151	100	428

Test coverage	**96,73 %**
Test successful coverage	**96,73 %**
Normal case	**41,36 %**
Abnormal case	**35,28 %**
Boundary case	**23,36 %**

Figure 4.1: Unit test case

Test Cases and Test Reports: Report5_Test Report_LockR.xlsx

TEST STATISTICS

Project Name	*Lock.R - Smart Locker System with Drone Delivery*	**Creator**	**binhtntse182370**	
Project Code	*LockR*	**Reviewer/Approver**	**huynqbse180211**	
Document Code	*LockR_Test Report_v1.0*	**Issue Date**		*09/11/2026*
Notes	*Release 1 includes 45 modules of the mobile app, kiosk, admin web, backend services and locker controller.*			

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Sign Up	8	0	0	0	8
2	Sign In with Password	8	0	0	0	8
3	Sign In with OTP or Google	6	0	0	0	6
4	Reset Password	6	0	0	0	6
5	Admin Login with 2FA	7	0	0	0	7
6	Kiosk Login by QR	6	0	0	0	6
7	Manage Profile	7	0	0	0	7
8	Search Stores and Lockers	7	0	0	0	7
9	Locker Layout Box Map	6	0	0	0	6
10	Send Parcel	7	1	0	0	8
11	Pay by VietQR SePay	7	0	0	0	7
12	Pay with Loyalty Points	5	0	0	0	5
13	Drop Parcel and Share Code	6	0	0	0	6
14	Change PIN and Delegate Pickup	6	0	0	0	6
15	Pick Up Parcel at Kiosk	6	1	0	0	7
16	Rent Locker	6	0	0	0	6
17	Extend and End Rental	6	0	0	0	6
18	Overdue Fee	6	0	0	0	6
19	My Orders and Incoming Orders	6	0	0	0	6
20	Rate Order and Complaint	6	0	0	0	6
21	Refund Request	6	0	0	0	6
22	Promotions and Vouchers	7	0	0	0	7
23	Loyalty Points	5	0	0	0	5
24	Report Locker Incident	6	0	0	0	6
25	Book Drone Delivery	8	0	0	0	8
26	Drone Delivery Tracking	6	0	1	0	7
27	Drone Home Pickup	6	0	0	0	6
28	Drone Dispatch and Loading	6	0	0	0	6
29	Drone Launch and Flight Failure	5	0	1	0	6
30	Drone Fleet Management	6	0	0	0	6
31	Technician Shift	5	0	0	0	5
32	Handle Incident Ticket	7	0	0	0	7
33	Periodic Inspection	6	0	0	0	6
34	Emergency Open and IoT Devices	5	1	0	0	6
35	Admin Dashboard and Revenue	6	0	0	0	6
36	Manage Users	5	1	0	0	6
37	Manage Stores Lockers Boxes	7	0	0	0	7
38	Manage Drones and Incidents	6	0	0	0	6
39	Process Refunds	5	0	0	0	5
40	Manage Promotions and Loyalty	5	0	0	0	5
41	Business Settings and Jobs	5	0	0	0	5
42	Notifications	5	0	1	0	6
43	Knowledge Base and AI Assistant	7	0	0	0	7
44	Bluetooth Open Test	5	0	0	0	5
45	Cabinet Controller	5	1	1	0	7
	Sub total	273	5	4	0	282

Test coverage **98,58** %
Test successful coverage **96,81** %

Figure 4.2: Test Statistic